SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

BCSE302P - DATABASE SYSTEMS LAB

Semester : Winter Semester 25-26 **Faculty : Dr.S.Geetha**

EXERCISE 5

Title: Single Row Functions

1. OBJECTIVE

- To understand and apply SQL single row (scalar) functions
- To use string, numeric, date, and NULL-handling functions in SQL queries
- To perform row-wise data manipulation and analysis

2. SINGLE ROW FUNCTIONS

Single row functions operate on each row individually and return one result per row.

Categories:

- String Functions
- Numeric Functions
- Date Functions
- Conversion Functions
- NULL Handling Functions

STUDENT TABLE

student_id	student_name	dept	age
101	Arun	CSE	20
102	Banu	ECE	21
103	Charan	CSE	19
104	Divya	MECH	22
105	Eshwar	CSE	21
106	Farah	CIVIL	20

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EMPLOYEE TABLE

emp_id	emp_name	dept	salary	age
1	Anand	IT	45000.75	28
2	Bhavya	HR	38000.40	32
3	Charan	IT	52000.60	26
4	Divya	FINANCE	60000.00	35
5	Ezhil	HR	40000.20	29
6	Farooq	IT	30000.90	24

ENROLLMENT TABLE

enroll_id	student_id	course_id	marks	semester
1	101	201	78	3
2	101	204	85	3
3	102	201	68	3
4	102	203	72	3
5	103	204	90	2
6	103	202	75	2
7	104	205	60	4
8	105	201	88	3
9	105	203	82	3
10	106	205	55	4

A. STRING FUNCTIONS

Functions: UPPER(), LOWER(), LENGTH(), SUBSTRING(), CONCAT(), TRIM()

1. Display student names in uppercase.
   SELECT UPPER(student_name) FROM STUDENT;
2. **Display employee names in lowercase.**
   SELECT LOWER(emp_name) FROM EMPLOYEE;
3. **Display length of each student name.**
   SELECT student_name, LENGTH(student_name) FROM STUDENT;
4. **Display first three characters of employee names.**
   SELECT SUBSTRING(emp_name,1,3) FROM EMPLOYEE;
5. **Concatenate student name and department.**
   SELECT CONCAT(student_name,' - ',dept) FROM STUDENT;

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B. NUMERIC FUNCTIONS

Functions: ROUND(), CEIL(), FLOOR(), ABS(), MOD()

6. Display employee salary rounded to nearest integer.

   SELECT emp_name, ROUND(salary) FROM EMPLOYEE;

7. Display salary divided by 3 and rounded to 2 decimals.

   SELECT emp_name, ROUND(salary/3,2) FROM EMPLOYEE;

8. Display absolute difference between salary and 50,000.

   SELECT emp_name, ABS(salary - 50000) FROM EMPLOYEE;

9. Display remainder when salary is divided by 5.

   SELECT emp_name, MOD(salary,5) FROM EMPLOYEE;

C. DATE FUNCTIONS

Functions: CURRENT_DATE, NOW(), DATEDIFF(), YEAR()

10. Display current system date.

    SELECT CURRENT_DATE;

11. Display employee experience in days.

    SELECT emp_id, DATEDIFF(CURRENT_DATE, join_date) FROM EMPLOYEE;

D. NULL HANDLING FUNCTIONS

Functions: IS NULL, IS NOT NULL, COALESCE(), NULLIF()

12. Display enrollment records where marks are NULL.

    SELECT * FROM ENROLLMENT WHERE marks IS NULL;

13. Display marks, replacing NULL with zero.

    SELECT enroll_id, COALESCE(marks,0) FROM ENROLLMENT;

14. Display NULL if marks are equal to zero.

    SELECT enroll_id, NULLIF(marks,0) FROM ENROLLMENT;

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7. CONDITIONAL FUNCTION – CASE

15. Display student result as Pass/Fail using CASE.

```
SELECT student_id,
CASE WHEN marks >= 50 THEN 'Pass' ELSE 'Fail' END AS Result
FROM ENROLLMENT;
```

3. EXPECTED OUTCOMES

After executing this experiment, the student will be able to:

- Apply single row functions for data formatting and manipulation
- Perform numeric and string-based computations
- Handle NULL values effectively
- Use conditional logic in SQL queries

4. EXPERIMENT QUESTIONS

What is a single row function?

Difference between single row and aggregate functions?

Purpose of COALESCE()?

Explain CASE with an example.

Can single row functions be used in WHERE clause? Why?

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5. PRACTICE DATABASE –

Use the same tables created and execute the following queries.

A. STRING FUNCTIONS

1. Display all student names in uppercase.
2. Display employee names in lowercase.
3. Find the length of each student name.
4. Display first three characters of employee names.
5. Concatenate student name and department.
6. Display employee names with trimmed spaces.
7. Display students whose names contain letter 'a'.
8. Display last two characters of student names.
9. Replace 'IT' with 'INFO' in employee department.
10. Display student names in reverse order.

B. NUMERIC FUNCTIONS (10 QUESTIONS)

1. Display employee salary rounded to nearest integer.
2. Display salary rounded to two decimal places.
3. Display highest absolute difference between salary and 50000.
4. Display remainder when salary is divided by 4.
5. Display ceiling value of salary divided by 7.
6. Display floor value of salary divided by 5.
7. Display salary increased by 10%.
8. Display salary difference from average salary.
9. Display employee salaries as positive values.
10. Display salary divided by age (rounded).

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C. DATE FUNCTIONS (10 QUESTIONS)

(Assume a column join_date exists if needed)

1. Display current system date.
2. Display current timestamp.
3. Display current year.
4. Display employees joined in the current year.
5. Calculate experience in days.
6. Extract month from current date.
7. Display employees joined before current date.
8. Display difference between two given dates.
9. Display week number of current date.
10. Add 30 days to current date.

D. NULL HANDLING FUNCTIONS (10 QUESTIONS)

1. Display enrollment records where marks are NULL.
2. Replace NULL marks with zero.
3. Replace NULL marks with average marks.
4. Display students whose marks are NOT NULL.
5. Display NULL if marks are equal to zero.
6. Count number of NULL marks.
7. Display marks after handling NULL using COALESCE.
8. Display students with missing evaluation.
9. Compare marks using NULL-safe conditions.
10. Display enrollment records with evaluated marks only.

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E. CONDITIONAL FUNCTIONS – CASE (10 QUESTIONS)

1. Display Pass/Fail based on marks.
2. Classify students as Distinction / First Class / Fail.
3. Display salary grade (High/Medium/Low).
4. Display semester category using CASE.
5. Assign bonus eligibility based on salary.
6. Display age group of employees.
7. Categorize marks as Excellent/Good/Average.
8. Display department type using CASE.
9. Convert numeric semester into text.
10. Display evaluation status using CASE.